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Electromyography Activity of Posterior Oblique Sling Muscles During High-Speed Running in Elite Soccer Players: Risk Factors for Hamstring Injury

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Electromyography activity of Posterior Oblique Sling muscles During high speed Running in elite soccer players: Risk factors for hamstring muscle injury

Abstract

Objective: This study examined how changing Posterior Oblique Sling muscles (POS) activity during the Prone Hip Extension (PHE) test can influence the activity of POS muscles and the risk of hamstring injury during high-speed running.

Method: Twenty-four male soccer players were divided into Non-Delayed (NDG, N=13) and Delayed groups (DG, N=11) (delay in activation of hamstring and gluteus maximus) based on activity patterns of POS muscles (Gluteus Maximus (GM), Hamstring muscles (H), and contralateral Lumbar Erector Spine (CLES)) during the PHE test. Then, they ran on a treadmill at 20 Km/h, and muscle activity of these muscles was recorded. Before recording muscle activity, we omitted participants with limited range of motion (ROM) to eliminate the effect of ROM on muscle activity patterns.

Results: Results showed that GM activities were significantly higher in the NDG at 7-10% of the running cycle, while the DG had greater GM activities at 55-62%. Semitendinosus (ST) activities were significantly higher in the DG at 45-70%, and the NDG exhibited greater ST activities at 95-100%. Biceps Femoris (BF) activities were significantly higher in the DG at 0-15% and 95-100%, while the NDG had higher BF activity at 45-62%. BF -to-GM activity ratio was significantly higher in the DG at 7-11%, and BF -to-Contralateral Lumbar Erector Spine (CLES) activity ratio was higher in the NDG at 52-55%.

Conclusion: Athletes in DG may have different POS muscle activity patterns compared to NDG, which may increase hamstring injury risk compared to the NDG due to observed activity patterns.

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Keywords: *hamstring, time-series, high-speed running, EMG*

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Introduction

Identifying modifiable risk factors could lead to developing injury prevention protocols linked with reduced injury rates [1, 2]. The research conducted on the risk of hamstring injury predominantly concentrated on factors related to muscle strength [3-5], neuromuscular adaptations after injury[6], and their role in injury recurrence [7, 8]. Within the last decades, musculoskeletal deficit treatment and prevention strategies have transitioned from emphasizing strength and flexibility to prioritizing motor control exercises [9-11]. Intra- and intermuscular coordination and their role in hamstring injury have been studied to exhibit neuromuscular coordination deficits [12-16]. It has been unanimously confirmed that neuromuscular inhibition and the property of intra-muscular coordination is linked to the hamstring injury [7, 12, 17, 18]. Nevertheless, these studies had limited study designs since they did not encompass underlying injury-related mechanisms or assess other local muscles. Besides, they all focused on the differences between injured and uninjured individuals, lacking the ability to establish cause-and-effect relationships between intra- and intermuscular coordination and hamstring injuries.

Along with the hamstrings, the gluteus maximus (GM) and the (superficial) lumbar erector spine (LES) generate and control hip and lumbar extensor torques (i.e., torques resisting external flexion moments) necessary for force closure and stabilization of the pelvic girdle. [19-25]. This function, where all these three muscles coordinate forces and control the joint, is called Posterior Oblique Sling (POS) [24, 26]. In addition, the magnitude and frequency of POS activity could effectively transfer power from the lower limb to the trunk and vice versa [26-31]. To evaluate the quality of motor control in the lumbopelvic region and the activity magnitude and pattern of POS, the Prone Hip Extension (PHE) test was used [19, 24]. By examining the activity pattern of the POS muscles, a new perspective on the balance of synergistic balance and muscle inhibition in the posterior region of the body could be gained [32]. Some studies showed

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a relationship between POS muscle activity during the PHE test and the risk of hamstring injury[32, 33].

In a study conducted on subjects with hamstring injuries, Emami, Arab [33] found a significant increase in the activity of GM and medial hamstring (MH) muscles in injured individuals. They stated that these changes are a compensatory mechanism for greater regional stability[33]. However, this article can't establish cause-and-effect relationships between POS muscle activity and hamstring injury. In a prospective study, Schuermans investigated the relationship between the activity pattern of the POS muscles in the PHE test and the risk of hamstring injury in soccer players. The results showed that in 9 injured athletes, the muscle activity pattern started with the Contralateral Lumbar Erector Spine (CLES) muscles, followed by the hamstring and the GM muscles. They concluded that the delay in hamstring and GM activity compared to the CLES muscles in the PHE test increases the risk of hamstring injury [32].

Although they showed the link between POS muscles activity pattern, in the PHE test, and hamstring injury, no study has explored how these changes in activity pattern contribute to hamstring injuries during high-speed running. Studying the effects of POS muscles activity patterns on hamstring injuries during high-speed running could reveal multi-dimensional factors causing it. Motor control patterns observed during low-load tasks such as the Prone Hip Extension (PHE) test are thought to reflect ingrained neuromuscular coordination strategies governed by central motor programs and feed-forward control. When these patterns become maladaptive—such as delayed gluteus maximus activation—they tend to persist under higher-load, dynamic conditions due to the CNS’s reliance on established activation sequences[12, 34]. Therefore, dysfunctional activation identified in the PHE test may mirror the altered Posterior Oblique Sling (POS) coordination that occurs during high-speed running, predisposing athletes to increased hamstring loading and injury risk. Additionally, existing literature mainly relies on the discrete analysis of electromyography, which gives vague information about the activity

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patterns throughout the running cycle. Application of time-series analysis over running cycles could provide in-depth information about POS muscle activity patterns and pinpoint any deficits throughout the cycle. To this end, this study aimed to determine how the activity patterns of POS muscles during the PHE test affect their electromyographic (the magnitude and ratio of muscle activity (time-series analysis)) activity during high-speed running, which is the most common mechanism of a hamstring injury, on the treadmill.

It was hypothesized that individuals with delayed GM and HM activation during the PHE test would exhibit altered POS coordination during high-speed running.

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Material and Methods

Study design

This study adopted a cross-sectional study design, where dependent variables were GM, H (ST and BF), and CLES muscles activity during high-speed running, and independent variables were muscle activation patterns during the PHE test. The tests were conducted in the biomechanics laboratory of the University of Tehran. The Institutional Review Board for the Protection of Human Subjects in Research at the University of Tehran ethically approved this study (Ethics code: IR.UT.SPORT.REC.1401.009.)

Participants

Twenty-four collegiate soccer players were divided into two groups based on their performance in the Prone Hip Extension (PHE) test: NDG (n = 13; age = 21.78 ± 3.42 years; height = 179.57 ± 0.07 cm; weight = 70.96 ± 8.09 kg; BMI = 21.94 ± 1.11 kg/m²) and DG (n = 11; age = 21.85 ± 2.93 years; height = 178.21 ± 0.04 cm; weight = 70.53 ± 6.01 kg; BMI = 22.17 ± 1.17 kg/m²). Based on previous studies, with statistical significance set at a two-sided level of 0.05, power of 0.8, and effect size 1.2 a minimum of 8 participants was required for each groups[8]. Although a priori power analysis indicated a minimum of eight participants per group, additional eligible individuals were available during recruitment. Therefore, the final sample included 11 participants in one group and 13 in the other, in accordance with the predefined inclusion and exclusion criteria. The inclusion criteria were a) no experiences of lower limbs and lower-back musculoskeletal (hamstring in particular) injuries within last 24 months[35], b) no visible and detectable postural abnormalities [36]and c) no muscle shortness and range of motion (ROM) limitations in the lumbopelvic and hip region according to the tests taken [32]. Before tests, participants were briefed on the study's objectives and the associated risks. Subsequently, they provided their consent by signing a written document acknowledging their informed decision to participate.

Procedure

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To eliminate other moderator variables (here ROM), we omitted participants with limited ROM (see ROM assessment section) [32]. Then, four ElectroMyoGraphy (EMG) (Myon Aktos, AG, Switzerland, sampling frequency = 2000Hz) surface electrodes (Medico, South korea, 27*37mm) were placed on the bulky volume of the GM, BF, and ST of the dominant leg and the CLES[32], following skin shaving, abrading, and cleaning (Figure 1). We selected the CLES, as force transmission across the pelvis occurs cross-coordinated, and this crossing POS also works synergistically in daily locomotion [37-41]. Following the identification and palpation of the targeted muscles in accordance with the SENIAM guidelines by a trained professional, the electrodes were affixed to the designated anatomical locations[42]. After checking the quality of the EMG signal in each of the four channels , three Maximal Voluntary Isometric Contractions (MVIC) trials were acquired per muscle. To assess muscle activation, maximal voluntary isometric contractions (MVICs) were obtained for normalization. For the lumbar erector spinae, participants performed a resisted back extension against force applied at the scapular level. Hamstring MVICs were recorded while participants resisted a knee extension torque at 30° flexion, with the lower leg supported by the tester. For the gluteus maximus, participants resisted hip flexion torque during hip extension. In each test, participants gradually increased effort to maximum within 3 s, maintained peak contraction for 5 s, and then relaxed.[32]. This procedure was conducted according to the SENIAM guidelines [42]. After that, the PHE test was employed to group participants based on their pattern of POS muscle activity (see PHE test section). After dividing participants into two groups, the high-speed running test was conducted on the treadmill (see running task section).

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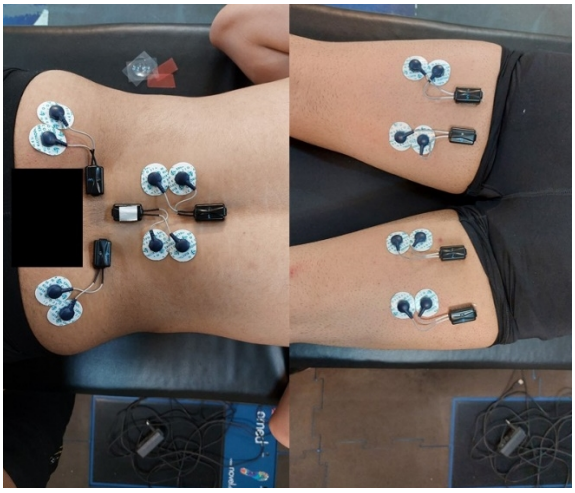

Figure 1. Electrode placement on the GM, CLES, BF, and ST.

ROM assessment

These clinical tests include hamstring flexibility (active knee extension while the hip is at a 90-degree angle)[43], the flexibility of the iliopsoas and rectus femoris muscles (modified Thomas test) [44], hip flexion ROM [45], internal and external rotation ROMs of the hip [45], lumbar flexion ROM (mSchober test) [46], and the ability to bear the tension of the posterior chain (slump test) [32] (Figure 2).

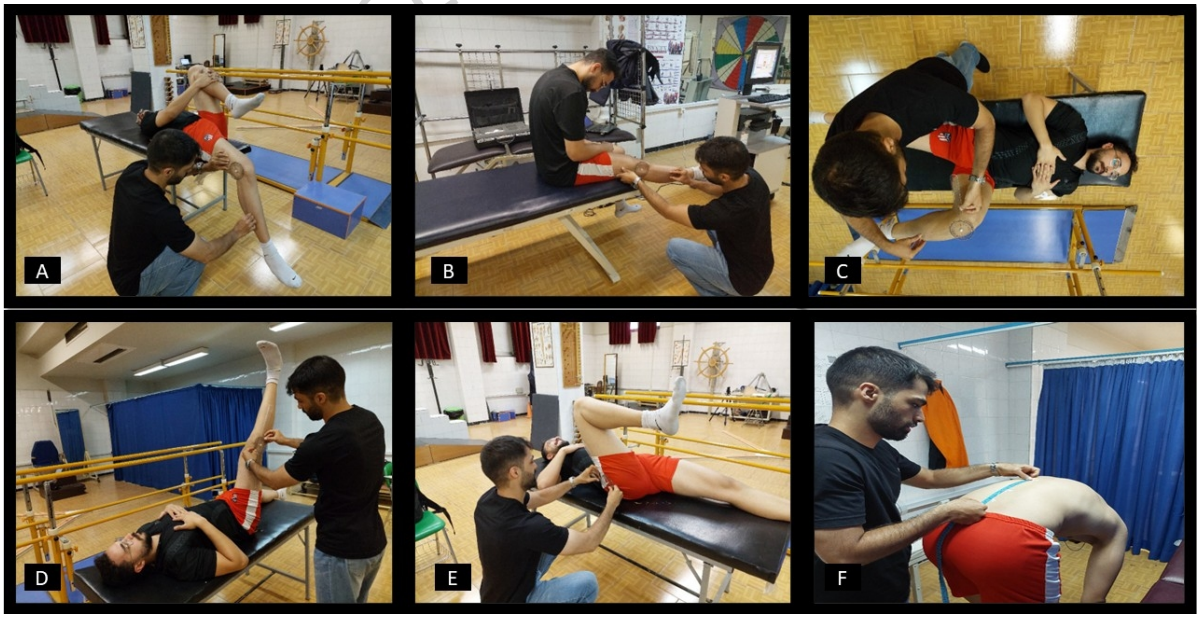

Figure 2. Mobility and ROM tests. A: modified Thomas test, B: slump test, C: internal and external rotation ROMs of the hip, D: active knee extension, E: hip flexion ROM, F: mSchober test

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PHE test

The PHE test evaluates the activity pattern of muscles involved in the hip extension movement. In this test, participants lay in a neutral prone position with the head down straight and both arms positioned next to their trunks (Figure 3). The test was performed on a dominant leg [32]. The dominant leg was defined as the participant's preferred kicking leg, consistent with previous research on lower-limb dominance in athletic populations[47]. Afterward, they were asked to lift the hip as much as desired while the knee was fully extended. This test was performed three times, and the activity pattern of the desired muscles was recorded. In this test, we averaged BF and ST's muscle activity and considered it HM muscle. Participants with GM and H activity onsets earlier than CLES were labeled as 'NDG'(N=13), while those whose CLES was activated earlier than either GM or HM were labeled as 'DG'(N=11) [24, 32, 48, 49].

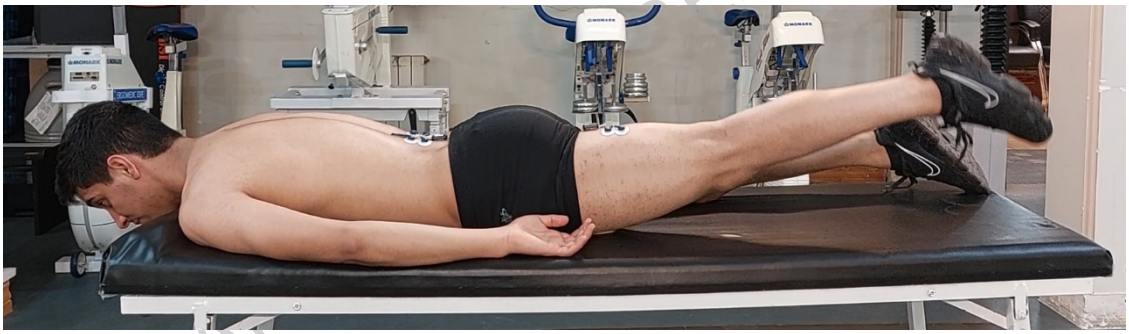

Figure 3. PHE test

Running task

The same muscles evaluated in the PHE test were also assessed during running, with the exception of separate analyses of BF and ST. Participants walked and ran slowly for 10 minutes during warm-up at their preferred speeds. Since hamstring injury occurs at high-speed running, participants ran at 20 km/h without an inclination on a treadmill [8]. An Inertial Measurement Unit (IMU) sensor (Myon, Switzerland, 32 x 17 x 12 mm, 2000Hz) was placed on the lower back (L5 level) participants throughout the running task [50]. Two bouts of 30 strides, inter-separated by 5 minutes of rest interval (to prevent fatigue), were recorded for each

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participant. 3D acceleration and EMG signals were recorded using Aktos EMG-made-easy software and exported for further analyses.

Data analysis

PHE

The recorded EMG signals were trimmed from two seconds before the PHE test start to 3 seconds after the hip was extended. Following a 20Hz high-pass filtering (4th-order Butterworth), signals were rectified in the 50-millisecond window for onset detection [51]. The muscle onset was identified when the amplitudes of the activity surpassed three standard deviations from the mean amplitude of the baseline [52]. The processed muscle activity onsets were put into the temporal algorithm to determine the sequence of muscle activity (see PHE test). (Table 1)

Table 1. Muscle activity patterns during the PHE test

Patterns	Delayed (D)/Non-delayed (ND)	Frequency
GM-HM-CLES	ND	5
HM-GM-CLES	ND	8
CLES-GM-HM	D	5
CLES-HM-GM	D	6

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Running

To detect each gait cycle during running, we employed the adjusted algorithm for gait event detection using IMU sensors (vertical acceleration signal) by Del Din, Godfrey [50]. To identify heel-strike events, an algorithm based on the inverted pendulum model was applied. This model has been shown to be a stable and valid framework for gait analysis across various walking speeds[50, 53, 54].. Since the present study required only the detection of heel contact with the ground, rather than complete stance or toe-off phases, this approach was deemed appropriate. The algorithm effectively identifies the heel-strike event, defined as the instant of initial contact within each running cycle. Using an IMU to assess this cycle

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presents some challenges in precisely defining its phases; therefore, we followed established frameworks. During high-speed running, the stance phase typically constitutes 25–30% and the swing phase 70–75% of the cycle. In this study, the early stance phase was defined as 0–10% and the late swing phase as 90–100% of the cycle to facilitate structured phase-based analysis [54].

Following the identification of gait events, EMG signals were trimmed from the initial contact of one leg to the subsequent initial contact of the same leg, thereby representing one complete cycle. The raw EMG signals are then bandpass filtered (cutoff frequency of 20 to 400 Hz) using a fourth-order Butterworth filter [51]. After that, to generate a linear envelope for each of the muscles during each running cycle, signals were full-wave rectified and low-pass filtered using a 2nd-order Butterworth filter (9Hz cutoff frequency). After that, they were divided by the MVIC values to calculate the activity percentage throughout each cycle [55]. Muscle activities were subsequently normalized to 101 data points for time-series analysis. For each participant, 20 running cycles were extracted and averaged to obtain a representative curve.

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Statistical analysis

The normality of data distribution was checked and confirmed using the Shapiro–Wilk statistical test. A independent t-test of the spm1d package (v0.4.3) (www.spm1d.org) was employed to compare lower limb muscle activity and activity ratio throughout running cycles between NDG and DG. The associated effect size was estimated using Cohen's d, and its magnitude was determined as follows: $d<0.2$ = small effect size, $0.2<d<0.8$=moderate effect size, and $d>0.8$=large effect size. The significance level was set at $\alpha=0.05$, and MATLAB software (version 2022a, MathWorks, Inc., Natick, MA, USA) was used to conduct all data and statistical analyses described above.

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Results

ROM result

No significant difference was observed between the mobility and flexibility of lumbopelvic region muscles between NDG and DG (table 2).

Table 2. Mobility and flexibility results of the groups

	NDG N=13	DG N=11	*p*	Cohen`s D
Iliopsoas flexibility (Degrees)	3 ± 2.42	2 ± 2.01	0.14	0.44
Rectus femoris flexibility (Degrees)	80 ± 2.23	81 ± 2.40	0.45	0.43
Active knee extension (Degrees)	88 ± 3.55	88 ± 4.72	0.97	0.01
Hip flexion ROM (Degrees)	120 ± 4.04	120 ± 4.65	0.95	0.02
Hip internal rotation ROM (degree)	32 ± 2.64	33 ± 2.13	0.24	0.41
Hip external rotation ROM (degree)	38 ± 4.05	39 ± 2.86	0.52	0.28
Knee extension during the slump (degree)	20 ± 2.92	19 ± 3.14	0.54	0.32
M`Schober test (cm)	6.49 ± 0.62	6.41 ± 0.62	0.69	0.12

EMG activity

GM activities were significantly higher among NDG at the beginning of the running cycle (7-10%, p=0.036, d>0.06) (Figure 4-A). Almost in the middle of the running cycle (55-62%), DG portrayed greater GM activities (p<0.001, d>0.14) (Figure 4-A). ST activities, on the other hand, were significantly higher among DG in the middle of the running cycle (45-70%, p<0.001, d>0.14) (Figure 4-B). On the contrary, NDG exhibited greater ST

activities in the final stages of the running cycle (95-100%, p =0.025, d>0.14).

The BF activities were significantly higher among DG at the beginning of the running cycle (0-15%, p<0.001, d>0.14) (Figure 4-C). Also, at the end of the running cycle, the BF showed higher activities in the DG (95-10%, p=0.034, d>0.06) (Figure 4-C). On the contrary, the BF activity was higher in the NDG in the middle of the running cycle (45-62%, p<0.001, d>0.14) (Figure 4-C). There was no significant difference in CLES activity between both groups. (P>0.05) (Figure 4-D).

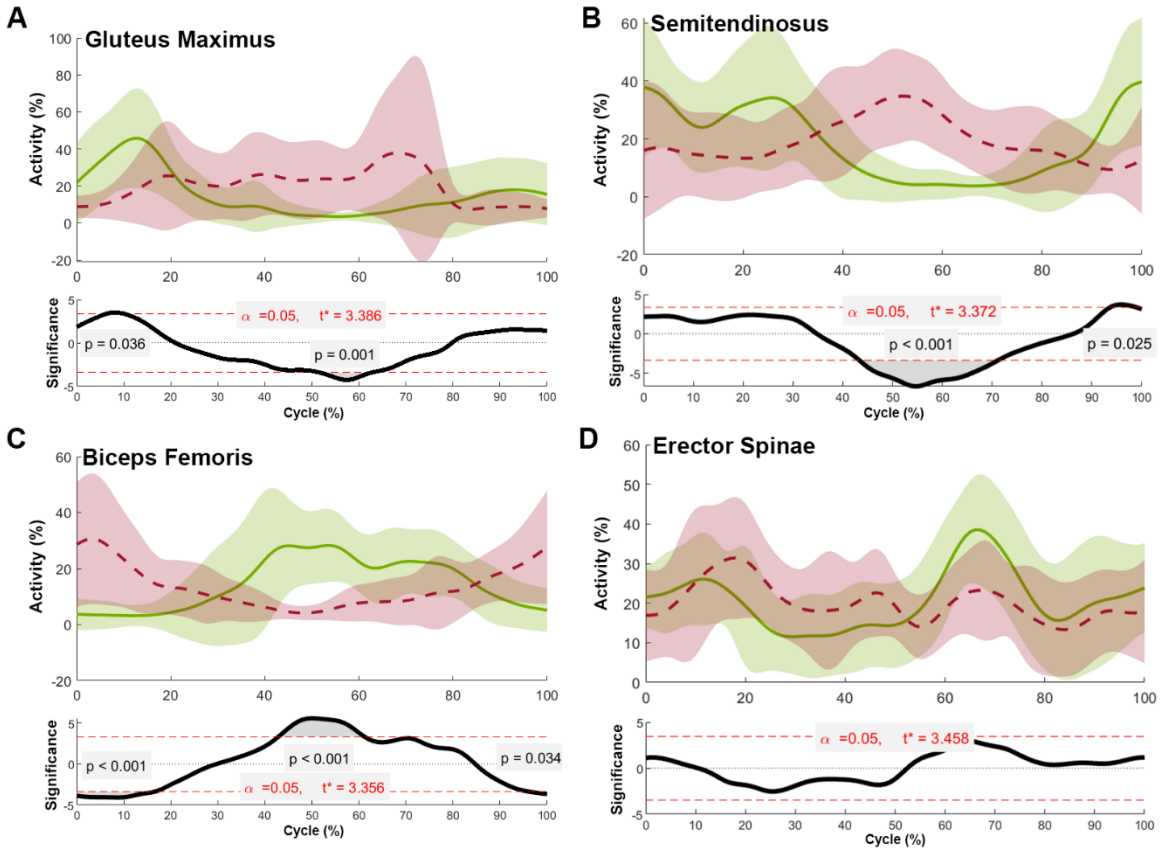

Figure 4. Electromyography activity of A) Gluteus Maximus, B) Semitendinosus, C) Biceps Femoris, and D) Erector Spinae among participants with DG (red dashes) and NDG (green lines). Significance differences (α<0.05) were highlighted below activity graphs.

BF-to-GM activity ratio was significantly higher in the DG at the beginning of the running cycle (7-11%, p=0.046, d>0.06) (Figure 5-A). In contrast, the BF-to-GM activity ratio was higher in the NDG in the middle

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of the running cycle (38-60%, $p<0.001$, d>0.14). BF-to-ES activity ratio was significantly higher in DG at the beginning of the running cycle (0-5%, p=0.039, d>0.06) (Figure 5-B), while the BF-to-ES activity ratio was higher in the NDG in the middle of the running cycle (52-55%, $p=0.047$, d>0.06).

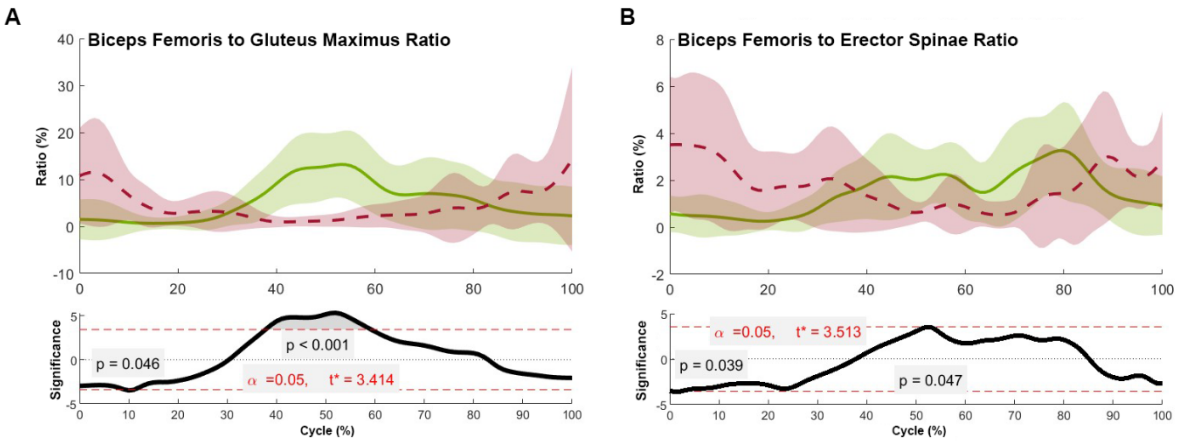

Figure 5. A) Biceps Femoris to Gluteus Maximus, and B) Biceps Femoris to Erector Spinae activity ratios among participants with DG (red dashes) and NDG (green lines). Significance differences (α<0.05) were highlighted below activity graphs.

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Discussion

This research investigated the role of muscle activity pattern changes in the POS muscle, as determined by the PHE test, on the electromyography activity of POS during high-speed running. This study showed that changes in POS muscle activity patterns can change POS muscle activity during running.

The results of our research showed that the activity of the gluteus maximus muscle in a running cycle is different in two groups. Muscles such as the GM and medius, which control the position of the hip and pelvis, have a potential effect on the length of the hamstring muscle[17, 56, 57]. It is important to note that the GM and internal and external abdominal oblique muscles can considerably reduce hamstring tension during running [56, 58]. The GM's proper activity (either magnitude or timing) during running could provide sufficient pelvis stability and help with a strong link between the upper- and lower limbs [13, 35]. The role of the gluteus maximus during running has been investigated by various studies [59-61]. Lieberman et al.'s (2006) analysis showed that the gluteus maximus plays a vital role in hip extension during the stance phase, which helps control trunk flexion during running [59]. Also, this research showed that the timing of this muscle activity also controls the trunk's flexion movement rate [59]. In addition, researchers showed that the activity of this muscle at the initial foot contact has its highest activity during the running cycle [59-61]. A prospective study by Scheuerman et al. (2017) showed that a 10% increase in GM activity during running reduces the risk of hamstring injury by 6 to 20% [13]. Results of our study portrayed that GM activities were significantly higher among NDG during the early stance phase, which is the most dangerous phase for HM injuries [62]. This indicates that NDG had proper support from GM, which may prevent excessive HM stretching [56, 63, 64].

Besides, BF-to-GM ratios were significantly higher in DG during the stance phase, portraying less GM support in this group than in their NDG peers. As synergistic entities, BF had to exert additional force to

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compensate for the deficiency in GM, whether to assist in managing applied forces on the structure (decelerating vertical mechanical loads) or to contribute to propulsive force (acceleration). This repetitive loading on the BF might increase the risk of overuse injuries throughout high-speed running [62].

DG, on the other side, portrayed significantly more GM activity during the forward swing phase (55-60%), which contradicts the results of the Schuermans, Danneels [13] study, where healthy participants exhibited increased GM activities[13]. However, a research study mentioned that GM was not particularly active in this part of the running cycle among healthy controls, and the BF showed more activity [18]. In our study, BF activity magnitudes at this phase of the running cycle were significantly higher in NDG, which is in line with the previously mentioned study [18]. To this end, we may claim that the considerably lower GM activities in the NDG do not indicate a deficit. Instead, the delayed GM activity in the DG signifies lower GM activity within this group.

Our results showed that the activity of the ST and BF in a running cycle is different in the two groups. The ST and BF muscles are activated in maximal eccentric activities in the swing phase [17, 18, 57]. These synergists work alternately in a complex process of neuromuscular coordination[17, 65, 66]. It has been reported that the ST exhibits the highest activity magnitudes during hamstring-related training programs and running, compared to the BF [65, 67-72]. Due to the natural morphologic structure of ST (longer muscle fibers), it has more capacity for stretching than BF[28]. This feature enables ST to absorb more eccentric stretching tensions and control the torques in the hip and knee joints during the late swing phase [18, 65, 66]. To this end, ST protects BF when mechanical stresses are applied to the lower limb in the later swing phase. In addition, although ST and BF have the same origin, each could undergo excessive mechanical stresses if their activity patterns (either spatial or temporal) are disrupted [12, 17, 18, 66, 70, 71]. The symmetrical activity pattern of these two muscles may indicate a compensatory and

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inconsistent neuromuscular coordination pattern, which may lead to inefficiency in the activity of these two muscle groups and ultimately causes fatigue[12]. This disrupted activity pattern may increase the risk of hamstring injury due to fatigue [12, 73]. Our results showed that if there is a delay in GM and HM activity during the PHE test, the activity pattern of the hamstring muscles (ST and BF) may will be completely incoherent. Despite this, in NDG, the BF muscle is more active in the middle of the swing phase, and its activity decreases in the late swing phase, which is in line with previously mentioned studies[18, 65, 66]. However, in the DG, the activity of the BF muscle increased in the late swing and early stance phases, where the ST muscle was supposed to be more active to absorb mechanical loads. In contrast, the ST was significantly more active during the mid-swing phase in DG, which might indicate two main concepts: first, to compensate for overall force production of HM muscles, and second, to prevent further BF fatigues[12, 28, 70]. Due to the shorter muscle length of the BF compared to the ST, its capacity to withstand eccentric tensions and control knee and hip torques is considerably lower [28]. As a result, any dysfunction in the ST muscle or coordination in these synergist muscles may contribute to injure both.

No significant difference was observed in the CLES muscles' activity magnitude between groups. The LES muscles activate biphasically during high-speed running [74, 75]. One of these activations is at initial foot contact, and the other is in the swing phase, just like the pattern shown in the present study. This activity pattern indicates that these muscles control trunk flexion torques during running [74-76]. Our result showed that BF-to-ES ratios were significantly higher in DG in the beginning and middle of the running cycle, portraying less ES activity magnitude in this group than in their NDG peers. The BF muscle may be more active in controlling the pelvis position due to disrupted activity of the lumbar ES muscle during running. This increased activity can overload the BF and may increase the risk of hamstring injury [8].

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Based on our results, sufficient and efficient synergistic activity in the entire POS is necessary to avoid excessive load on the HM muscles during locomotion. Suitable synergistic activity in the POS may guarantees homeostasis and may contribute to prevent excessive load on the hamstring muscle while running. Compensatory activity patterns may put the muscles at greater risk of injury. Delays in GM and the dominant activity of the BF over the ST may lead to excessive eccentric loading on BF and make it prone to injury.

The present study had some limitations. First, in ideal conditions, all variables should have been measured in the acceleration phase, not in constant speed. In this phase, the load and stress on the hamstrings are much higher than when running at a constant speed. Second, Although treadmill running provides a controlled environment for measuring muscle activation, it differs biomechanically from overground running. Studies have shown that treadmill running is associated with reduced hamstring activity, particularly during the late swing phase, due to lower horizontal braking forces and altered stride mechanics. Additionally, treadmill running typically produces smaller vertical ground reaction forces and reduced hip extension angles compared to overground running[77, 78]. These differences may partly explain variations in posterior chain muscle activation, suggesting that future research should replicate the current findings under overground, sport-specific conditions. Nevertheless, this study had strengths compared to other studies. To our knowledge, no research has divided people based on the pattern of muscle activity, which is very similar to the pattern of walking and running, and the role of the pattern of intermuscular activity in the risk of hamstring injury has not been investigated. Additionally, we excluded athletes with limited lumbopelvic region and hip ROMs to focus solely on POS activity patterns. Besides, very few studies have examined the risk of hamstring injury during high-speed running.

Conclusion

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The present study showed that delay in HM and GM activity in the PHE test may increase the risk of hamstring injury during high-speed running by disrupting the activity pattern of POS muscles. Based on the present findings, targeted motor-control interventions should focus on retraining the temporal and synergistic activation patterns of the Posterior Oblique Sling (POS). Exercises that emphasize early and coordinated gluteus maximus activation relative to the hamstrings may correct delayed activation patterns observed in the Delayed Group. Effective examples include bridging variations (e.g., single-leg or band-resisted bridges emphasizing gluteal pre-activation) and bird-dog exercises with contralateral limb loading, which enhance cross-sling neuromuscular coordination between the gluteus maximus and contralateral lumbar erector spinae. Furthermore, reactive neuromuscular training using resistance bands or unstable surfaces can improve intermuscular timing and dynamic pelvic stability under load[79-82]. Incorporating these exercises into running-specific rehabilitation may optimize lumbopelvic control, reduce eccentric overload on the biceps femoris, and mitigate the risk of recurrent hamstring injury.

Clinical Relevance

This study shows that delayed gluteus maximus and hamstring activation in the Posterior Oblique Sling (POS) disrupts neuromuscular coordination during high-speed running, potentially increasing hamstring injury risk. The Prone Hip Extension (PHE) test can identify dysfunctional POS patterns, and targeted motor control training to optimize activation sequencing of the gluteus maximus, hamstrings, and lumbar erector spinae may enhance lumbopelvic stability and reduce injury risk. These findings support POS-focused neuromuscular retraining in athlete prevention and rehabilitation programs.

Declarations

Abbreviation

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POS: Posterior Oblique Sling, PHE: Prone Hip Extension, DG: Delayed Group, NDG: None Delayed Group, GM: Gluteus Maximus, H: Hamstring, CLES: Contralateral Lumbar Erector Spine, LES: Lumbar Erector Spine, BF: Biceps Femoris, ST: SemiTendinosus, MH: Medial Hamstring, ROM: Range Of Motion, MVC: Maximum Voluntary Contraction, MVIC: Maximum Voluntary Isometric Contraction, SENIAM: Surface Electromyography for the Non-Invasive Assessment of Muscles, EMG: ElectroMyoGraphy, IMU: Inertial Measurement Unit, SPM: Statistical Parametric Mapping

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and was ethically approved by the Institutional Review Board for the Protection of Human Subjects in Research at the University of Tehran (Ethics code: IR.UT.SPORT.REC.1401.009). Written informed consent was obtained from all participants.

Clinical trial number

not applicable.

Consent for publication

Written informed consent for participant information and images to be published was provided by the participants.

Availability of data and materials

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare no competing interests.

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Authors' contributions

All of the authors conceived of the presented idea. MS, HM, ES, and MH developed the theory and performed the computations. JS verified and did the analytical methods. HM and ES encouraged MS to investigate and supervise the findings of this work. All authors discussed the results and contributed to the final manuscript.

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References

1. Emery, C.A., *Injury prevention and future research.* Epidemiology of pediatric sports injuries: team sports, 2005. **49**: p. 170-191.
2. Ter Stege, M.H., et al., *Effect of interventions on potential, modifiable risk factors for knee injury in team ball sports: a systematic review.* Sports Medicine, 2014. **44**: p. 1403-1426.
3. Pellicer-Chenoll, M., et al., *Comparison of conventional hamstring/quadriceps ratio between genders in level-matched soccer players.* Revista andaluza de medicina del deporte, 2017. **10**(1): p. 14-18.
4. Dauty, M., P. Menu, and A. Fouasson-Chailloux, *Cutoffs of isokinetic strength ratio and hamstring strain prediction in professional soccer players.* Scandinavian journal of medicine & science in sports, 2018. **28**(1): p. 276-281.
5. Silder, A., D.G. Thelen, and B.C. Heiderscheit, *Effects of prior hamstring strain injury on strength, flexibility, and running mechanics.* Clinical Biomechanics, 2010. **25**(7): p. 681-686.
6. Areia, C., et al., *Neuromuscular changes in football players with previous hamstring injury.* 2019. **69**: p. 115-119.
7. Fyfe, J.J., et al., *The role of neuromuscular inhibition in hamstring strain injury recurrence.* Journal of Electromyography and Kinesiology, 2013. **23**(3): p. 523-530.
8. Daly, C., et al., *The biomechanics of running in athletes with previous hamstring injury: A case-control study.* Scandinavian journal of medicine & science in sports, 2016. **26**(4): p. 413-420.
9. Bourne, M.N., et al., *An evidence-based framework for strengthening exercises to prevent hamstring injury.* Sports Medicine, 2018. **48**: p. 251-267.
10. Huygaerts, S., et al., *Mechanisms of hamstring strain injury: interactions between fatigue, muscle activation and function.* Sports, 2020. **8**(5): p. 65.
11. Shield, A.J. and M.N. Bourne, *Hamstring injury prevention practices in elite sport: evidence for eccentric strength vs. lumbo-pelvic training.* Sports medicine, 2018. **48**: p. 513-524.
12. Schuermans, J., et al., *Biceps femoris and semitendinosus—teammates or competitors? New insights into hamstring injury mechanisms in male football players: a muscle functional MRI study.* British journal of sports medicine, 2014. **48**(22): p. 1599-1606.
13. Schuermans, J., et al., *Proximal neuromuscular control protects against hamstring injuries in male soccer players: a prospective study with electromyography time-series analysis during maximal sprinting.* The American journal of sports medicine, 2017. **45**(6): p. 1315-1325.
14. Bourne, M.N., et al., *Hamstring and gluteal activation during high-speed overground running: impact of prior strain injury.* Journal of Sports Sciences, 2021. **39**(18): p. 2073-2079.
15. Van Der Made, A.D., et al., *Intramuscular tendon injury is not associated with an increased hamstring reinjury rate within 12 months after return to play.* British journal of sports medicine, 2018. **52**(19): p. 1261-1266.
16. Schuermans, J., et al., *Susceptibility to hamstring injuries in soccer: a prospective study using muscle functional magnetic resonance imaging.* The American journal of sports medicine, 2016. **44**(5): p. 1276-1285.
17. Higashihara, A., et al., *Functional Differences between Individual Hamstring Muscles at Different Running Speeds: 1800: Board# 237 June 2 3: 30 PM-5: 00 PM.* Medicine & Science in Sports & Exercise, 2010. **42**(5): p. 404.

ARTICLE IN PRESS

ARTICLE IN PRESS

18. Higashihara, A., et al., *Functional differences in the activity of the hamstring muscles with increasing running speed.* Journal of sports sciences, 2010. **28**(10): p. 1085-1092.
19. Arab, A.M., et al., *Altered muscular activation during prone hip extension in women with and without low back pain.* Chiropractic & manual therapies, 2011. **19**(1): p. 1-6.
20. Bruno, P., *The importance of diagnostic test parameters in the interpretation of clinical test findings: The Prone Hip Extension Test as an example.* The Journal of the Canadian Chiropractic Association, 2011. **55**(2): p. 69.
21. Bruno, P.A. and J. Bagust, *An investigation into motor pattern differences used during prone hip extension between subjects with and without low back pain.* Clinical Chiropractic, 2007. **10**(2): p. 68-80.
22. Chance-Larsen, K., C. Littlewood, and A. Garth, *Prone hip extension with lower abdominal hollowing improves the relative timing of gluteus maximus activation in relation to biceps femoris.* Manual Therapy, 2010. **15**(1): p. 61-65.
23. Guimarães, C.Q., et al., *Electromyographic activity during active prone hip extension did not discriminate individuals with and without low back pain.* Brazilian Journal of Physical Therapy, 2010. **14**: p. 351-357.
24. Frank, C., P. Page, and R. Lardner, *Assessment and treatment of muscle imbalance: The Janda approach: Human kinetics.* Search in, 2009.
25. Janda, V., C. Frank, and C. Liebenson, *Evaluation of muscular imbalance.* Rehabilitation of the spine: a practitioner's manual, 1996. **6**: p. 97-112.
26. Vleeming, A., G.-j. Kleinrensink, and R. Stoeckart, *The role of the hamstrings in pelvic and spinal function.* Movement, Stability & Low Back Pain, 1997: p. 207-210.
27. Pel, J., et al., *Biomechanical model study of pelvic belt influence on muscle and ligament forces.* Journal of biomechanics, 2008. **41**(9): p. 1878-1884.
28. Kellis, E., *Intra-and inter-muscular variations in hamstring architecture and mechanics and their implications for injury: a narrative review.* Sports Medicine, 2018. **48**(10): p. 2271-2283.
29. Sichting, F., et al., *Pelvic belt effects on sacroiliac joint ligaments: a computational approach to understand therapeutic effects of pelvic belts.* Pain physician, 2014. **17**(1): p. 43.
30. Vleeming, A. and R. Stoeckart, *The role of the pelvic girdle in coupling the spine and the legs: a clinical-anatomical perspective on pelvic stability.* Movement, stability & lumbopelvic pain, 2007: p. 113-137.
31. Vleeming, A., R. Stoeckart, and C. Snijders, *The sacrotuberous ligament: a conceptual approach to its dynamic role in stabilizing the sacroiliac joint.* Clinical Biomechanics, 1989. **4**(4): p. 201-203.
32. Schuermans, J., D. Van Tiggelen, and E. Witvrouw, *Prone hip extension muscle recruitment is associated with hamstring injury risk in amateur soccer.* International journal of sports medicine, 2017. **38**(09): p. 696-706.
33. Emami, M., A.M. Arab, and L. Ghamkhar, *The activity pattern of the lumbo-pelvic muscles during prone hip extension in athletes with and without hamstring strain injury.* International journal of sports physical therapy, 2014. **9**(3): p. 312.
34. Hodges, P.W. and K.J.P. Tucker, *Moving differently in pain: a new theory to explain the adaptation to pain.* 2011. **152**(3): p. S90-S98.
35. Smith, M.M.F., et al., *Gluteus medius activation during running is a risk factor for season hamstring injuries in elite footballers.* Journal of science and medicine in sport, 2017. **20**(2): p. 159-163.

ARTICLE IN PRESS

ARTICLE IN PRESS

36. Lun, V., et al., *Relation between running injury and static lower limb alignment in recreational runners.* British journal of sports medicine, 2004. **38**(5): p. 576-580.
37. Oh, J.-S., *Effects of pelvic belt on hip extensor muscle EMG activity during prone hip extension in females with chronic low back pain.* Journal of Physical Therapy Science, 2014. **26**(7): p. 1023-1024.
38. Oh, J.-S., et al., *Effects of performing an abdominal drawing-in maneuver during prone hip extension exercises on hip and back extensor muscle activity and amount of anterior pelvic tilt.* Journal of orthopaedic & sports physical therapy, 2007. **37**(6): p. 320-324.
39. Opar, D.A., et al., *Rate of torque and electromyographic development during anticipated eccentric contraction is lower in previously strained hamstrings.* The American journal of sports medicine, 2013. **41**(1): p. 116-125.
40. Tateuchi, H., et al., *Pelvic instability and trunk and hip muscle recruitment patterns in patients with total hip arthroplasty.* Journal of Electromyography and Kinesiology, 2013. **23**(1): p. 151-158.
41. Bahr, R., *Why screening tests to predict injury do not work—and probably never will...: a critical review.* British journal of sports medicine, 2016. **50**(13): p. 776-780.
42. Hermens, H.J., et al., *European recommendations for surface electromyography.* Roessingh research and development, 1999. **8**(2): p. 13-54.
43. Shamsi, M., M. Mirzaei, and S.S. Khabiri, *Universal goniometer and electro-goniometer intra-examiner reliability in measuring the knee range of motion during active knee extension test in patients with chronic low back pain with short hamstring muscle.* BMC Sports Science, Medicine and Rehabilitation, 2019. **11**(1): p. 1-5.
44. Clapis, P.A., S.M. Davis, and R.O. Davis, *Reliability of inclinometer and goniometric measurements of hip extension flexibility using the modified Thomas test.* Physiotherapy theory and practice, 2008. **24**(2): p. 135-141.
45. Nussbaumer, S., et al., *Validity and test-retest reliability of manual goniometers for measuring passive hip range of motion in femoroacetabular impingement patients.* BMC musculoskeletal disorders, 2010. **11**: p. 1-11.
46. Robinson, H.S. and A.M. Mengshoel, *Assessments of lumbar flexion range of motion: intertester reliability and concurrent validity of 2 commonly used clinical tests.* Spine, 2014. **39**(4): p. E270-E275.
47. van Melick, N., et al., *How to determine leg dominance: The agreement between self-reported and observed performance in healthy adults.* 2017. **12**(12): p. e0189876.
48. Sahrmann, S., *Concepts and principles of movement.* Diagnosis and treatment of movement impairment syndromes, 2001.
49. Wattananon, P., et al., *Lumbar Multifidus and Erector Spinae Muscle Synergies in Patients with Nonspecific Low Back Pain During Prone Hip Extension: A Cross-sectional Study.* PM&R, 2019. **11**(7): p. 694-702.
50. Del Din, S., A. Godfrey, and L. Rochester, *Validation of an accelerometer to quantify a comprehensive battery of gait characteristics in healthy older adults and Parkinson's disease: toward clinical and at home use.* IEEE journal of biomedical and health informatics, 2015. **20**(3): p. 838-847.
51. Sarvestan, J., et al., *The effect of ankle Kinesio™ taping on ankle joint biomechanics during unilateral balance status among collegiate athletes with chronic ankle sprain.* Physical Therapy in Sport, 2020. **45**: p. 161-167.

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52. Allison, G., *Trunk muscle onset detection technique for EMG signals with ECG artefact.* Journal of Electromyography and Kinesiology, 2003. **13**(3): p. 209-216.
53. Patoz, A., et al., *A single sacral-mounted inertial measurement unit to estimate peak vertical ground reaction force, contact time, and flight time in running.* 2022. **22**(3): p. 784.
54. Sibson, B.E., et al., *Using inertial measurement units to estimate spine joint kinematics and kinetics during walking and running.* 2024. **14**(1): p. 234.
55. Shiavi, R., C. Frigo, and A. Pedotti, *Electromyographic signals during gait: criteria for envelope filtering and number of strides.* Medical and Biological Engineering and Computing, 1998. **36**: p. 171-178.
56. Chumanov, E.S., B.C. Heiderscheit, and D.G. Thelen, *The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of sprinting.* Journal of biomechanics, 2007. **40**(16): p. 3555-3562.
57. Higashihara, A., et al., *Effects of forward trunk lean on hamstring muscle kinematics during sprinting.* Journal of sports sciences, 2015. **33**(13): p. 1366-1375.
58. Kumazaki, T., Y. Ehara, and T. Sakai, *Anatomy and physiology of hamstring injury.* International journal of sports medicine, 2012. **33**(12): p. 950-954.
59. Lieberman, D.E., et al., *The human gluteus maximus and its role in running.* Journal of Experimental Biology, 2006. **209**(11): p. 2143-2155.
60. Ellis, R.G., B.J. Sumner, and R. Kram, *Muscle contributions to propulsion and braking during walking and running: insight from external force perturbations.* Gait & posture, 2014. **40**(4): p. 594-599.
61. Willson, J.D., et al., *Male and female gluteal muscle activity and lower extremity kinematics during running.* Clinical biomechanics, 2012. **27**(10): p. 1052-1057.
62. Chumanov, E.S., B.C. Heiderscheit, and D.G. Thelen, *Hamstring musculotendon dynamics during stance and swing phases of high speed running.* Medicine and science in sports and exercise, 2011. **43**(3): p. 525.
63. Chumanov, E.S., et al., *Changes in muscle activation patterns when running step rate is increased.* Gait & posture, 2012. **36**(2): p. 231-235.
64. Heiderscheit, B.C., et al., *Hamstring strain injuries: recommendations for diagnosis, rehabilitation, and injury prevention.* journal of orthopaedic & sports physical therapy, 2010. **40**(2): p. 67-81.
65. Schache, A.G., et al., *Stretch and activation of the human biarticular hamstrings across a range of running speeds.* European journal of applied physiology, 2013. **113**: p. 2813-2828.
66. Onishi, H., et al., *EMG-angle relationship of the hamstring muscles during maximum knee flexion.* Journal of Electromyography and Kinesiology, 2002. **12**(5): p. 399-406.
67. Oliver, G.D. and C.P. Dougherty, *Comparison of hamstring and gluteus muscles electromyographic activity while performing the razor curl vs. the traditional prone hamstring curl.* The Journal of Strength & Conditioning Research, 2009. **23**(8): p. 2250-2255.
68. Mendiguchia, J., et al., *Nonuniform changes in MRI measurements of the thigh muscles after two hamstring strengthening exercises.* The Journal of Strength & Conditioning Research, 2013. **27**(3): p. 574-581.
69. Mendiguchia, J., et al., *The use of MRI to evaluate posterior thigh muscle activity and damage during Nordic hamstring exercise.* The Journal of Strength & Conditioning Research, 2013. **27**(12): p. 3426-3435.
70. McAllister, M.J., et al., *Muscle activation during various hamstring exercises.* The Journal of Strength & Conditioning Research, 2014. **28**(6): p. 1573-1580.

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71. Orishimo, K.F. and M.P. McHugh, *Effect of an eccentrically biased hamstring strengthening home program on knee flexor strength and the length-tension relationship.* The Journal of Strength & Conditioning Research, 2015. **29**(3): p. 772-778.
72. Schache, A.G., et al., *Mechanics of the human hamstring muscles during sprinting.* Medicine & science in sports & exercise, 2012. **44**(4): p. 647-658.
73. Lee, J.W., et al., *Eccentric hamstring strength deficit and poor hamstring-to-quadriceps ratio are risk factors for hamstring strain injury in football: A prospective study of 146 professional players.* Journal of science and medicine in sport, 2018. **21**(8): p. 789-793.
74. Cappellini, G., et al., *Motor patterns in human walking and running.* Journal of neurophysiology, 2006. **95**(6): p. 3426-3437.
75. Neumann, D.A., *Kinesiology of the musculoskeletal system-e-book: foundations for rehabilitation*. 2016: Elsevier Health Sciences.
76. Saunders, S.W., et al., *Changes in three dimensional lumbo-pelvic kinematics and trunk muscle activity with speed and mode of locomotion.* Clinical biomechanics, 2005. **20**(8): p. 784-793.
77. Van Hooren, B., et al., *Is motorized treadmill running biomechanically comparable to overground running? A systematic review and meta-analysis of cross-over studies.* 2020. **50**(4): p. 785-813.
78. Schache, A.G., et al., *Lower-limb muscular strategies for increasing running speed.* 2014. **44**(10): p. 813-824.
79. Distefano, L.J., et al., *Gluteal muscle activation during common therapeutic exercises.* 2009. **39**(7): p. 532-540.
80. Selkowitz, D.M., et al., *Which exercises target the gluteal muscles while minimizing activation of the tensor fascia lata? Electromyographic assessment using fine-wire electrodes.* 2013. **43**(2): p. 54-64.
81. Jang, T.-J., I.-C.J.P.T. Jeon, and Practice, *Effects of pelvic tilt control using visual biofeedback on gluteus maximus, multifidus, and hamstring activities during three different bridge exercises.* 2024. **40**(12): p. 2795-2804.
82. Yoon, J.-O., et al., *Effect of modified bridge exercise on trunk muscle activity in healthy adults: a cross sectional study.* 2018. **22**(2): p. 161-167.

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